

Project Design Phase-I Solution Architecture

Date	07 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture connects the **insurance fraud problem with a technological solution using machine learning techniques**. It defines how different components of the system work together to detect fraudulent insurance claims effectively. The main goals of the solution architecture are:

- **Collect and manage insurance claim data** from different sources such as policy records, customer details, and claim information.
- **Process and analyze the data** using data preprocessing techniques like cleaning, normalization, and feature selection.
- **Apply machine learning algorithms** to identify patterns and predict whether an insurance claim is fraudulent or genuine.
- **Provide a user-friendly interface** where users can enter claim details and get fraud prediction results.
- **Ensure efficient system performance** by enabling accurate fraud detection, reducing manual work, and improving decision-making for insurance companies

Example - Solution Architecture Diagram:

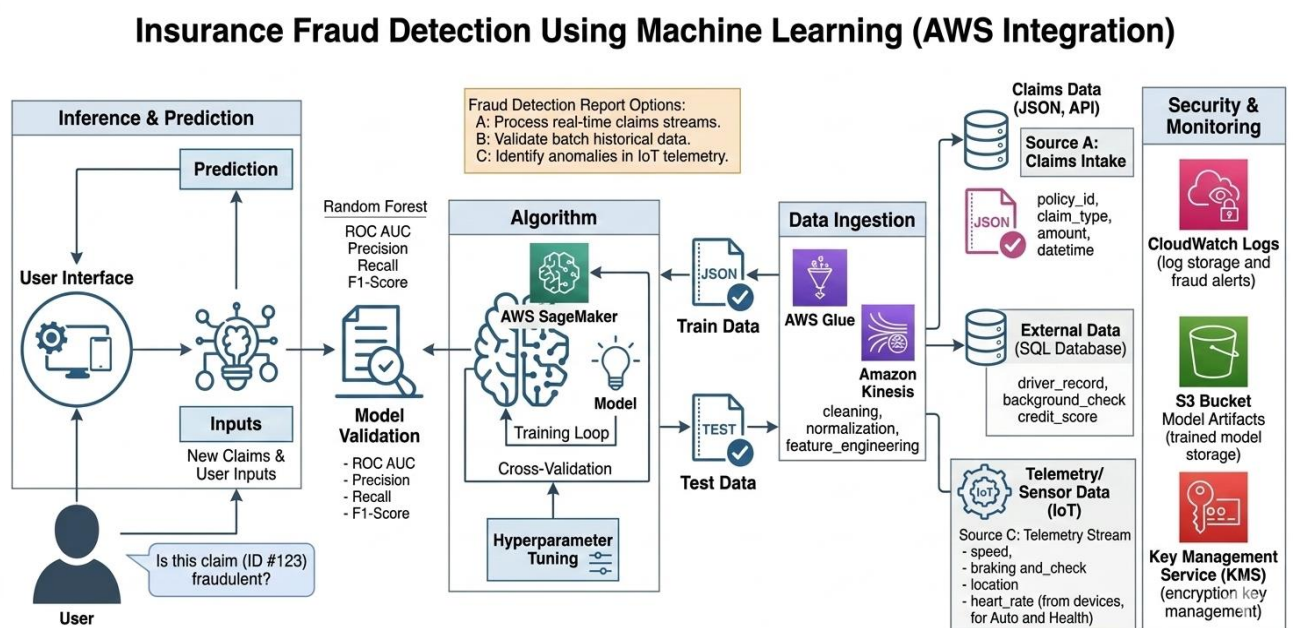


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-researchpowered-by-ai-on-aws-part-1-architecture-and-design-considerations/>